2. Binary Search

Binary Search works on sorted data by repeatedly dividing the search interval in half. It compares the target value to the middle element and eliminates half of the remaining elements based on the comparison.

```
def binary_search(data, target):
    low = 0
    high = len(data) - 1
    while low <= high:
        mid = (low + high) // 2
        if data[mid] == target:
                return mid  # Return index of the target element
        elif data[mid] < target:
                low = mid + 1
        else:
                high = mid - 1
    return -1  # Return -1 if the target is not found
```

Explanation:

- The function repeatedly divides the list into halves, adjusting the search boundaries (`low` and `high`) based on the comparison with the middle element.
